COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

XINZE LI

Abstract. The Almgren–Pitts combinatorial argument is a recurring technical step in min-max constructions of embedded minimal hypersurfaces: a cover-refinement on the parameter interval, followed by a bookkeeping argument that turns local near-critical savings into a global supremum improvement. Each author re-derives it in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g) with $H_n(M;\mathbb{Z}) = 0$ (for instance $M = S^{n+1}$), direct minimization of the n-dimensional Hausdorff measure over integral currents yields only the empty hypersurface; a non-trivial minimal hypersurface must be constructed as a saddle-type critical point of the area functional. The construction, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* scheme; regularity of the output is due to Pitts [Pit81] in ambient dimension up to 7 and to Schoen–Simon [SS81] in higher dimensions.

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set* $\operatorname{Sing}\Sigma$ *has Hausdorff dimension at most $n-7$.*

A key step in Pitts's proof, recurring in [CDL03, DLT13, DLR18, DL16, MN14], is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Theorem 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integer rectifiable varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Theorem 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

The combinatorial core of the proof of the Claim, which is the precise theorem we formalize, reads as follows.

Date: April 24, 2026.
Correspondence: `lixinze@math.utoronto.ca`.

Theorem 1.3 (Combinatorial core)**.** *Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open neighborhood $U_t \ni t$ and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exist a finite family of closed pieces $\{p_l \subseteq [0,1]\}_{l\in\iota}$, continuous replacement energies $\{E_l : [0,1] \to \mathbb{R}\}_{l\in\iota}$, and savings $\{s_l \geq 1/(4N)\}_{l\in\iota}$ satisfying:*

(I) *$f - E_l \geq s_l$ on p_l;*
(II) *every $t \in [0,1]$ belongs to at most two of the pieces p_l;*
(III) *$\{t \in [0,1] : f(t) \geq m_0 - 1/N\} \subseteq \bigcup_l p_l$.*

Remark 1.4. The non-trivial content of Theorem 1.3 is the *construction* of the cover $\{p_l\}$ from unstructured witness data: a Lebesgue-number argument on the compact near-critical set produces an initial cover satisfying the skip-2 spacing condition (a) of [DLT13]; a bounded smallest-index induction refines it to an injective piece-to-interval assignment; and a parity argument on the spacing condition delivers the two-fold overlap (II). The bookkeeping corollary below (the existence of a competitor f' with reduced supremum) is a three-line arithmetic consequence.

Remark 1.5 (Lean scaffolding not load-bearing)**.** The Lean implementation ✓ `exists_refinement` carries additional scaffolding not exposed in the statement of Theorem 1.3: a pseudo-metric ambient space X with a `PairableCover X` typeclass (Definition 2.1), and each local witness carries an abstract paired cover c_t in X (the `cover` field of `LocalWitness`, Definition 2.2). None of the `PairableCover` fields (`leftRegion`, `rightRegion`, `diameter`, `regions_disjoint`, `diameter_nesting`) nor the `cover` field is consumed by the proof of the combinatorial core; they are retained as extension hooks for future GMT applications in which E_t arises from a geometric replacement family on X. The ✓ `exists_sup_reduction` signature additionally carries a vestigial $m_0 > 0$ hypothesis (marked `_hm_pos` to flag it as unused) inherited from an earlier design. The hypotheses and conclusion of Theorem 1.3 above reflect the load-bearing mathematical content; see `docs/design-notes.md §§12--13` for the pending scaffolding resolution scheduled for v0.2.

Translation from the De Lellis–Tasnady setting to Theorem 1.3: the boundary $\partial\Omega_t$ becomes a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; and the almost-minimizing hypothesis, which in the original proof is the input to Lemma 3.1 of [DLT13], is absorbed into the existence of the local witness (the caller must supply witnesses wherever they would be produced in the ambient GMT setting). The output of Theorem 1.3 is the combinatorially assembled cover data $\{(p_l, E_l, s_l)\}$; the modified sweepout in the original proof is the geometric lift of the scalar competitor produced by the corollary below.

Corollary 1.6 (Sup reduction)**.** *Under the hypotheses of Theorem 1.3, there exists $f' : [0,1] \to \mathbb{R}$ with*

$$\sup f' \;\leq\; m_0 - \frac{1}{4N}.$$

The bookkeeping underlying Corollary 1.6 is strictly more general than the one-parameter statement it certifies: the same arithmetic applies to any compact, nonempty topological space K with continuous $f : K \to \mathbb{R}$ and any scalars (δ, ε) with $0 < \varepsilon \leq \delta$, taking as input any finite cover of the δ-super-level set of f with per-piece savings $\geq \varepsilon$ and at-most-two overlap. The Lean library ships both forms (Corollaries 2.10 and 2.11 in §2.3); Theorem 1.3 is the combinatorial construction that turns local-witness data into such a cover in the 1D case.

Both results reappear in §2: Theorem 1.3 as the library's main theorem (Theorem 2.8, §2.2), and Corollary 1.6 as the one-parameter sup-reduction corollary (Corollary 2.11, §2.3).

Scope. This note formalizes the proof of the Claim (the combinatorial argument on $[0,1]$) as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the

$(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships one main theorem (the combinatorial core, §2.2) on $[0, 1]$, together with two sup-reduction corollaries (§2.3): a generic compact-K form and its one-parameter specialization. The formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

In the proof of Theorem 1.2, the Lemma 3.1 of [DLT13] supplies, for each near-critical parameter t, an $\mathcal{H}^n$-disjoint pair of open sets $(U'_{1,t}, U'_{2,t})$ in M^{n+1} inside which a local replacement is performed. We abstract this pair as an instance of a class, stripped of its GMT content.

Definition 2.1 (`PairableCover` class). [✓ `PairableCover`]

A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

The replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ at a near-critical t has its Hausdorff measure decreased by at least $1/(4N)$ inside U'_t. We keep only the scalar *replacement energy* $E : K \to \mathbb{R}$ and the uniform saving inequality; the geometric family is discarded.

Definition 2.2 (`LocalWitness` structure). [✓ `LocalWitness`]

A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

The proof of the Claim works on the set $K \subset [0, 1]$ of parameters at which the sweepout is within $1/N$ of the min-max value. We use the same convention.

Definition 2.3 (Near-critical set). [✓ `nearCritical`]

For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

At the end of the refinement the data is a family $\{\overline{J_l}\}$ covering K, with each J_l carrying a replacement whose energy saving is at least $1/(4N)$ and with two-fold overlap. We abstract this output: the near-critical threshold $1/N$ becomes a parameter δ and the saving floor $1/(4N)$ a parameter $\varepsilon \leq \delta$.

Definition 2.4 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]

A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

The skip-2 spacing condition (a) of the original proof is a purely one-dimensional geometric condition on a family of open intervals. We isolate it in its own structure so the disjointness lemmas that depend only on spacing can be proved independently of the witness data.

Definition 2.5 (`SkippedSpacedIntervals` structure)**.** [✓ `SkippedSpacedIntervals`]

A `SkippedSpacedIntervals` records a natural number n, centers $c_1, \ldots, c_n \in \mathbf{I}$, positive radii $r_1, \ldots, r_n > 0$, and the skip-2 spacing axiom $c_i + r_i < c_{i+2} - r_{i+2}$ whenever $i + 2 \leq n$. The i-th open interval $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ is an abbreviation on this structure.

On top of the geometric structure, the initial cover also carries a witness center $w_i \in \mathrm{NC}$ with an attached local witness for each interval, together with the containment of each I_i inside the corresponding witness neighborhood. In our formalization the interval center c_i and the witness center w_i are recorded separately: grid points need not themselves lie in NC.

Definition 2.6 (`InitialCover` structure)**.** [✓ `InitialCover`]

Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries the data of a `SkippedSpacedIntervals` (Definition 2.5, satisfying spacing condition (a)), plus, for each $i = 1, \ldots, n$, a witness center $w_i \in \mathrm{NC}$ and a local witness at w_i with saving $1/(4N)$, subject to (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.

The refinement is described as an iteration: each I_j with $j \geq 2$ is either kept whole or split into two halves. The state of this iteration after ℓ moves is the following.

Definition 2.7 (`PartialRefinement` structure)**.** [✓ `PartialRefinement`]

Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. **Public API: combinatorial core.** The library's main theorem is the existence of a refined multiplicity-bounded cover of the near-critical set carrying the witness savings: the non-trivial combinatorial content extracted from the original proof. Its hypotheses coincide with the intro statement (Theorem 1.3); the output is a `FiniteCoverWithWitnesses` (Definition 2.4).

Theorem 2.8 (Combinatorial core)**.** [✓ `exists_refinement`]

Let $f : \mathbf{I} \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose every $t \in \mathrm{NC}$ admits a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.2). Then there exists a `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ (Definition 2.4): a finite family of closed pieces $\{p_l \subseteq \mathbf{I}\}$, continuous replacement energies $\{E_l : \mathbf{I} \to \mathbb{R}\}$, and savings $\{s_l \geq 1/(4N)\}$ satisfying conditions (I)–(IV).

The Lean signature of `exists_refinement` carries an ambient pseudo-metric X and a `PairableCover X` instance via `LocalWitness`; neither is consumed by the proof of Theorem 2.8. See Remark 1.5.

The proof is given in §3 as three chained stages: Stage A builds an initial cover from a Lebesgue-number argument on NC; Stage B runs a bounded smallest-index induction to a terminal partial refinement; Stage C assembles the output by extending the open-neighborhood saving bound to the closure via continuity and deriving two-fold overlap via the parity-gap disjointness argument.

2.3. **Public API: sup-reduction corollaries.** The arithmetic that turns a `FiniteCoverWithWitnesses` into a competitor with reduced supremum is a short scalar lemma independent of covers, pieces, or multiplicity: any competitor that saves a fixed amount on the relevant super-level set yields the same amount of supremum reduction.

Lemma 2.9 (Scalar sup reduction)**.** [✓ `sup_le_of_saving`]

Let K be a nonempty set and $f : K \to \mathbb{R}$ a bounded function with $m_0 = \sup f$. Let $g : K \to \mathbb{R}$ satisfy

(i) *$g(t) \leq f(t)$ for every $t \in K$;*
(ii) *$f(t) - g(t) \geq \varepsilon$ whenever $f(t) \geq m_0 - \delta$,*

where $0 < \varepsilon \le \delta$. *Then* $\sup g \le m_0 - \varepsilon$.

Proof. Fix $t \in K$. If $f(t) \ge m_0 - \delta$, then (ii) gives $g(t) \le f(t) - \varepsilon \le m_0 - \varepsilon$. If $f(t) < m_0 - \delta$, then (i) gives $g(t) \le f(t) < m_0 - \delta \le m_0 - \varepsilon$. Taking the supremum over t completes the proof. □

Lemma 2.9 is a general statement about lowering the supremum of a bounded real-valued function by exhibiting a competitor with a pointwise saving on the super-level set $\{f \ge m_0 - \delta\}$; no sweepout, cover, or multiplicity hypothesis enters. It applies verbatim whenever a pointwise saving can be established on the "where it matters" super-level set. The Lean library ships two corollaries of Theorem 2.8 via this lemma: the generic-K form (any `FiniteCoverWithWitnesses`), and the one-parameter specialization to $\mathbf{I}$ that chains through Theorem 2.8.

Corollary 2.10 (Sup reduction, generic form)**.** [✓ `exists_sup_reduction_of_cover`]
Let K be a compact, nonempty topological space, $f : K \to \mathbb{R}$ continuous with $m_0 = \sup f$, and $0 < \varepsilon \le \delta$. Given a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ *(Definition 2.4), there exists $f' : K \to \mathbb{R}$ with*

$$\sup f' \;\le\; m_0 - \varepsilon.$$

Proof. Set $f'(t) := f(t) - \sum_{l\,:\,t \in p_l} s_l$ and verify the hypotheses of Lemma 2.9. Condition (i): each $s_l > 0$, so $f'(t) \le f(t)$. Condition (ii): suppose $f(t) \ge m_0 - \delta$. By (IV) of Definition 2.4, $t \in p_l$ for some l; by (III), $s_l \ge \varepsilon$. The remaining savings in the sum (at most one more, by the two-fold overlap (II)) are non-negative, so $f(t) - f'(t) \ge \varepsilon$. The lemma yields $\sup f' \le m_0 - \varepsilon$. □

Corollary 2.11 (One-parameter sup reduction)**.** [✓ `exists_sup_reduction`]
Under the hypotheses of Theorem 2.8, there exists $f' : \mathbf{I} \to \mathbb{R}$ with

$$\sup f' \;\le\; m_0 - \tfrac{1}{4N}.$$

Corollary 2.11 is Corollary 2.10 applied to the `FiniteCoverWithWitnesses` produced by Theorem 2.8, with $(\delta, \varepsilon) = (1/N, 1/(4N))$.

2.4. **Internal lemmas.** The declarations below are the sub-lemmas of Theorem 2.8; proofs are sketched in §3.

Proposition 2.12 (Compactness of NC)**.** [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]
If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- ✓ `exists_initialCover` : existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` : existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` : at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` : the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.

The assembly itself—packaging the Stage A–B–C output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$—is Theorem 2.8 itself, whose Lean declaration ✓ `exists_refinement` chains the sub-lemmas above.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and

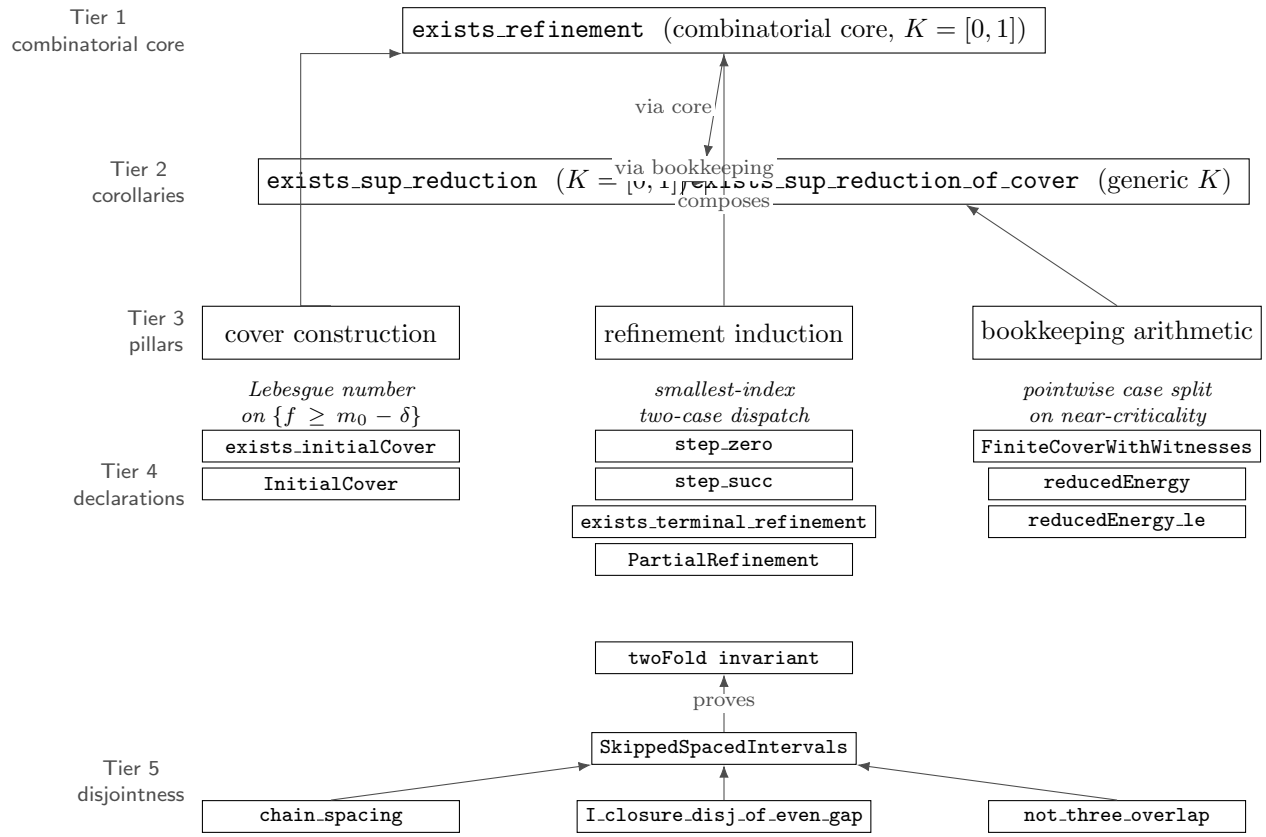

FIGURE 1. `CombArg` v0.1 architecture. The combinatorial core (Tier 1) is the main theorem `exists_refinement`: the 1D cover-refinement producing a `FiniteCoverWithWitnesses` from local witnesses on NC. Two sup-reduction corollaries (Tier 2) follow by scalar bookkeeping: the generic-K form `exists_sup_reduction_of_cover` and the one-parameter specialization `exists_sup_reduction`. Tier 1 is supplied by pillars P1 (cover construction) and P2 (refinement induction); the generic corollary is supplied by pillar P3 (bookkeeping arithmetic). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) is an abstract `SkippedSpacedIntervals` structure whose three lemmas combine into the two-fold overlap invariant consumed by Tier 1.

deriving TwoFold. The sup-reduction corollaries (Corollary 2.10 and Corollary 2.11) follow by Lemma 2.9 applied to the resulting cover; we sketch that 3-line argument after Stage C.

3.1. **Stage A: existence of the initial cover.**

Proposition 3.1 (Existence of an initial cover)**.** [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.6) exists.

Proof. Compactness of NC (Proposition 2.12) plus the open cover $\{U_t\}_{t\in\mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \ldots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c)

holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. **Stage B: the refinement induction.** Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case)**.** [✓ `step_zero`]
step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3 (Inductive step)**.** [✓ `step_succ_at`]
Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \dots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.7 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[1]

Proposition 3.4 (Termination)**.** [✓ `exists_terminal_refinement`]
Starting from step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Proof. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr′ with $i^* \notin$ rem(pr′) and rem(pr′) $\subseteq$ rem(pr)$\setminus\{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell+1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. **Stage C: assembly.** Stage C packages the refined covering as a `FiniteCoverWithWitnesses` (Definition 2.4), which the core theorem turns into the sup-reducing f'. We trace through [DLT13, §3.2, Step 2] in sequence, indicating at each step where the content lands in the formalization.

De Lellis and Tasnady begin by applying Lemma 3.1 to produce, at each near-critical parameter t, a replacement family $\{\partial\Omega_{i,t}\}$ with three properties: $\Omega_{i,t} = \Omega_t$ outside (a_i, b_i) and agrees with Ω_t outside U_i' inside (a_i, b_i); the global upper bound $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) + \frac{1}{4N}$; and, most importantly, the saving $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) - \frac{1}{2N}$ when $t \in (a_i + \delta, b_i - \delta)$. In our formalization this data is what a `LocalWitness` (Definition 2.2) carries: the scalar $E_{\sigma(k)} : K \to \mathbb{R}$ plays the role of $\mathcal{H}^n(\partial\Omega_{i,t})$, the neighborhood U_t plays (a_i, b_i), and the two separate bounds ("saving $-\frac{1}{2N}$ on the inner core" and "no-loss $+\frac{1}{4N}$ outside") collapse to a single uniform saving $\varepsilon = \frac{1}{4N}$ on U_t.[2] Stage B has already handed us pieces p_l on which this witness saving is valid (Proposition 3.6), so by Stage C we have per-piece scalar savings $s_l \ge \varepsilon$ directly.

The next observation in the proof is that if $t \in (a_i, b_i) \cap (a_j, b_j)$, then $j = i + 1$, and moreover $\mathrm{dist}(U_i', U_{i+1}') > 0$. This is the two-fold overlap of the refined covering, and in our formalization it is the `twoFold` field of `FiniteCoverWithWitnesses`, derived from the parity-gap spacing by Proposition 3.5. The parity-gap lemmas themselves live on the abstract `SkippedSpacedIntervals` (Definition 2.5) and are therefore independent of the witness data: `InitialCover` provides access to them by projecting out the underlying geometry.

The modified sweepout $\{\partial\Omega_t'\}$ is then defined piecewise: $\Omega_t' = \Omega_t$ outside all J_i; $\Omega_t' = \Omega_{i,t}$ if t lies in a single J_i; and on $J_i \cap J_{i+1}$ the sweepout blends the two replacements, $\Omega_t' = [\Omega_t \setminus (U_i' \cup U_{i+1}')] \cup$

[1] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

[2] This is strictly stronger than the original two-bound statement: the local witness asserts a uniform saving of $\frac{1}{4N}$ on all of U_t, whereas De Lellis–Tasnady only require a larger saving $\frac{1}{2N}$ on the inner core $(a_i + \delta, b_i - \delta)$ and bounded loss $\frac{1}{4N}$ elsewhere. The stronger abstract assumption removes the δ parameter of the inner core from the scalar theorem; the caller must verify the uniform version holds in the ambient setting.

$[\Omega_{i,t} \cap U_i'] \cup [\Omega_{i+1,t} \cap U_{i+1}']$. Stripped of the geometry this is additive scalar subtraction of savings over the pieces containing t: our reduced energy is

$$f'(t) \;=\; f(t) \;-\; \sum_{l\,:\,t\in\overline{J_l}} s_l,$$

and the two-fold invariant bounds the sum length by 2, matching the two-piece blend on overlaps.

The area accounting then splits on whether t is near-critical. For $t \notin K$, the point lies in at most two J_i's and each replacement can add at most $\frac{1}{4N}$, so

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) + \tfrac{1}{2N}.$$

For $t \in K$, the point lies in at least one inner core $(a_i + \delta, b_i - \delta)$, where the replacement saves $\ge \frac{1}{2N}$ in U_i', and in at most one other J_l where it costs $\le \frac{1}{4N}$ in U_l', for net

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) - \tfrac{1}{4N}.$$

In our formalization the same case analysis is the scalar argument for $f'(t) \le m_0 - \varepsilon$ inside the abstract core (the lemma `reducedEnergy_le` from §2.2): split on whether t lies in any piece of the cover, use the saving floor $s_l \ge \varepsilon$ in the positive branch and the covering condition $\{f \ge m_0 - \delta\} \subseteq \bigcup_l p_l$ in the negative branch. With $(\delta, \varepsilon) = (1/N, 1/(4N))$ this specializes to the arithmetic of the original proof; the abstract core reads off the same algebra for arbitrary (δ, ε) with $\varepsilon \le \delta$.

The rest of this subsection establishes the two propositions that Stage B leaves open: the saving bound extends from open J_k to $\overline{J_k}$ (continuity), and the two-fold invariant holds (a parity argument on the spacing condition (a)).

Proposition 3.5 (TwoFold via parity). [✓ `terminal_twoFold`]

For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \ldots, \overline{J_L}$.*

Proof. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i-j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \ldots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \ldots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound). [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \ge 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Proof. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \ge 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\ge$-inequalities under limits. □

Proof of Theorem 2.8. Chain Propositions 3.1 and 3.4: the former produces an `InitialCover` from the witness hypothesis, and the latter yields a terminal `PartialRefinement` pr of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i. Assemble the `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ by taking pieces $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and E_k the replacement energy of the $\sigma(k)$-th local witness. Invariant (I) is Proposition 3.6; invariant (II) is Proposition 3.5; invariant (III) holds by construction ($s_k = 1/(4N)$); coverage of NC chains as $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Corollary 2.10 then follows by the three-line bookkeeping proved in §2.3 applied to this `FiniteCoverWithWitnesses`; Corollary 2.11 is the specialization to $(\delta, \varepsilon) = (1/N, 1/(4N))$.

References

[Pit81] J. T. Pitts, *Existence and regularity of minimal surfaces on Riemannian manifolds*, Mathematical Notes 27, Princeton University Press, 1981.

[SS81] R. Schoen and L. Simon, *Regularity of stable minimal hypersurfaces*, Comm. Pure Appl. Math. 34 (1981), no. 6, 741–797.

[CDL03] T. H. Colding and C. De Lellis, *The min-max construction of minimal surfaces*, Surv. Differ. Geom. VIII (2003), 75–107.

[DLT13] C. De Lellis and D. Tasnady, *The existence of embedded minimal hypersurfaces*, J. Differential Geom. 95 (2013), no. 3, 355–388.

[DLR18] C. De Lellis and J. Ramic, *Min-max theory for minimal hypersurfaces with boundary*, Ann. Inst. Fourier 68 (2018), no. 5, 1909–1986.

[DL16] C. De Lellis, exposition of the combinatorial step in the min-max construction, Ann. Inst. Fourier, 2016. (Precise article reference to be confirmed against the author's library.)

[MN14] F. Marques and A. Neves, *Min-max theory and the Willmore conjecture*, Ann. of Math. (2) 179 (2014), no. 2, 683–782.